

X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation

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Abstract—Arsenic-induced skin disease presents a significant diagnostic challenge, particularly in resource-limited regions. While large deep learning models achieve high accuracy, their computational cost hinders practical deployment. This paper introduces X-ArsenicDistill, a novel framework that addresses this gap by employing knowledge distillation to create a lightweight yet powerful diagnostic model. We leverage a high-performance DeiT-Base model as a teacher to train a compact DeiT-Tiny student, using a composite loss function that transfers knowledge from the teacher’s soft targets and intermediate feature representations. Our large teacher model establishes a strong benchmark, achieving a test accuracy of 98.97%. Through our distillation framework, the lightweight student model achieves an impressive accuracy of 95.88%, demonstrating a notable improvement over the same model trained from scratch. The resulting student model is 15.5× smaller than its teacher, making it highly suitable for deployment on resource-constrained devices. Explainable AI analysis further confirms that the student successfully learns to focus on clinically relevant features, enhancing its trustworthiness for medical applications.

Index Terms—Knowledge Distillation, Arsenicosis, Skin Disease Classification, Vision Transformer, Medical Image Analysis, Explainable AI

I. INTRODUCTION

Arsenic poisoning of potable water is a serious social health emergency especially in South Asia whereby Bangladesh and India are hit the hardest with millions of people depending on tube wells as a source of drinking water [1]. Chronic exposure to this toxic metalloid causes arsenicosis which is a chronic disease the most noticeable symptom of which is dermatological lesions [2]. These lesions may precede serious health conditions such as skin cancer, and therefore, their early and proper diagnosis is a clinical requirement. Diagnosis, however, is frequently complicated by the morphological resemblance of these lesions to other frequent dermatological diseases. This creates a major problem to the health care providers particularly in the resource constrained environment.

Deep learning is an effective medical image analyzer which has been extensively used in dermatology in the past

decade [3]. CNN models, specifically, have demonstrated impressive levels of efficiency, being able to reach equal or even higher accuracy levels than trained dermatologists in detecting various skin diseases [4]. Vision Transformers (ViTs) have demonstrated more recently to have tremendous potential, in that they are able to extract global contextual information in an image which is potentially vital in identifying the slightest differences in lesions that are important to distinguish [5]. The models provide a way through to design automated, available, and precise diagnostic media that can complement clinical decision-making.

Despite the great progress that has been made, the use of advanced models especially transformer architecture on arsenic related skin diseases is yet to be fully explored. Although large-scale models are highly accurate, they have large computational and memory demands, which makes their application difficult in the highly resource-constrained settings to which they are the most appropriate. This presents an urgent requirement of light and efficient but generally accurate models.

To cope with this problem, the proposed paper presents a new framework, called X-ArsenicDistill, which is also founded on knowledge distillation [6]. We use a high-performance Data-efficient Image Transformer (DeiT-Base) [7] model as instructing a large model to train a smaller and efficient DeiT-Tiny model [7]. We cause the teacher to transfer into the student model the rich dark knowledge such as its subtle predictions and intermediate features representations which lead to higher performance than the student model would have achieved in isolation. With this approach, we can develop a model that is very accurate and computationally viable to be deployed in real-life situations.

The most important findings of this work can be outlined as follows:

- The design of a multi-faceted knowledge distillation framework, named X-ArsenicDistill that would be adaptable to the creation of efficient and precise models that

would be useful in the classification of arsenic-induced skin disease.

- We will prove that our distilled DeiT-Tiny student is much more effective than a baseline model trained with no parameter and performs at the level of state-of-the-art (SOTA) models despite being trained with a reduction in the count of its parameters by factor of $15.5\times$
- A qualitative examination of Attention Rollout [8] to offer insight into the model, indicating that our student model has been trained to attend towards clinically significant parts, and it is following the behavior of its significantly larger instructor.

II. RELATED WORKS

Deep learning in the detection of skin disease caused by arsenic is an expanding area of application, which mirrors general AI trends in dermatology. The first attempts were aimed at using pre-trained CNN architectures to compare the performance on this classification task.

Saha et al. [9] provided a detailed comparative study of widely used deep-learning models on the ArsenicSkinImageBD dataset, including the VGG-19, Resnet-50, MobileNetV2, and EfficientNet versions. They found EfficientNet-B1 to be the best with 94.69% testing accuracy through systematic fine-tuning. The study has laid an important foundation on the performance baselines and proved the viability of transfer learning in this clinical issue. Continuing these findings, Jobayer et al. [10] did not only increase classification to five classes (arsenic-induced lesions and melanoma and psoriasis), but also five classes were used. They introduced ARS-CNNs, a modified CNN with self-attention mechanisms to learn intricate, scattered patterns of skin disease and have 91% accuracy on their multi-class data. Model fusion and hybrid architecture were investigated by other researchers. Mehedi et al. [11] proposed ArsenicNet, which is a combination of pre-trained Xception backbone and Inception modules to take advantage of the strong feature extraction and multi-scale processing. This mixed method achieved an accuracy of 97.69% on binary classification, which is an act of synergy in architecture. The elusive goal of computational efficiency resulted in advanced lightweight models. Sintheia et al. [12] suggested an innovative lightweight hybrid CNN with dual-path architecture, one of them being Dense and Residual blocks, and the other one being Fire modules. Their model combined the parallel path features and had a 98.33% accuracy, which demonstrates strong and resource-efficient one-to-one design. On the same note, Hossen and Mondal [13] developed scalable solutions with custom lightweight CNN, with focus on comparison to transformer-based models such as ViT and Swin Transformer. Their smooth architecture had a 98.75% testing accuracy, which was competitive to larger counterparts at relatively low computation overhead requirements.

The literature shows a distinct trend of getting standard pre-trained models to highly specialized custom architectures. Although such design types expanded the limits of accuracy

and efficiency, they usually demand quite a bit of experimentation on the part of the architect. To complement our work, as opposed to coming up with new complex architectures, we consider knowledge distillation to endow simple and standard lightweight models (DeiT-Tiny) with the advanced knowledge of powerful teacher models (DeiT-Base). This little explored method of detecting arsenicosis presents a good strategy in an ideal performance-efficiency level.

III. METHODOLOGY

This section outlined the overall methodology of our proposed framework, X-ArsenicDistill, which will facilitate the effective and precise classification of the skin lesions of arsenic origin. The Framework shown in Fig. 1 represents the aggregate Framework. Our first step involves the description of the dataset itself and the preprocessing methods that were used to preserve the data quality and promote the reliability of the model. We then present the teacher-student architecture which is the center of our approach. Then we discuss the composite knowledge distillation loss function [6], which has a disaggregated form of its components. Lastly, we look at the way explainable AI approaches are integrated to provide transparent and understandable information about the decision-making of the model.

A. Dataset and Preprocessing

Our research is based on the publicly available set of dermatological images, namely ArsenicSkinImageBD [14], which deals with the skin diseases caused by arsenic exposure. The data set consists of 1,287 images which are categorized into two groups i.e. Infected (741 images) and Not Infected (546 images). This is a realistic challenge of this inherent class imbalance which our framework is meant to solve. The photographs were taken with a range of consumer-level smartphones in multiple villages in Bangladesh, which generates natural differences in lighting, scale, and perspective, which are useful to educate a generalizable model.

Each image is processed using a standardized preprocessing pipeline and then inputted into the models. During the preprocessing step, each image is downsampled to a standard resolution of 224×224 pixels to meet the requirements of the Vision Transformer input. The dataset is then split into training (70%), validation (15%), and testing (15%) subsets, where the proportions of classes are balanced across all of them. In order to increase the amount of training data and reduce the overfitting, a sequence of on-the-fly transformations are used. This can be done by geometric augmentations, such as random horizontal and vertical flips and rotations, and photometric augmentations, such as color jittering. Further enhancement of model robustness is also done using a Gaussian blur which is stochastic.

B. Model Architecture

X-ArsenicDistill is based on a paradigm of teacher-student that utilizes the Data-efficient Image Transformer (DeiT) architecture [7]. This model of knowledge distillation involves

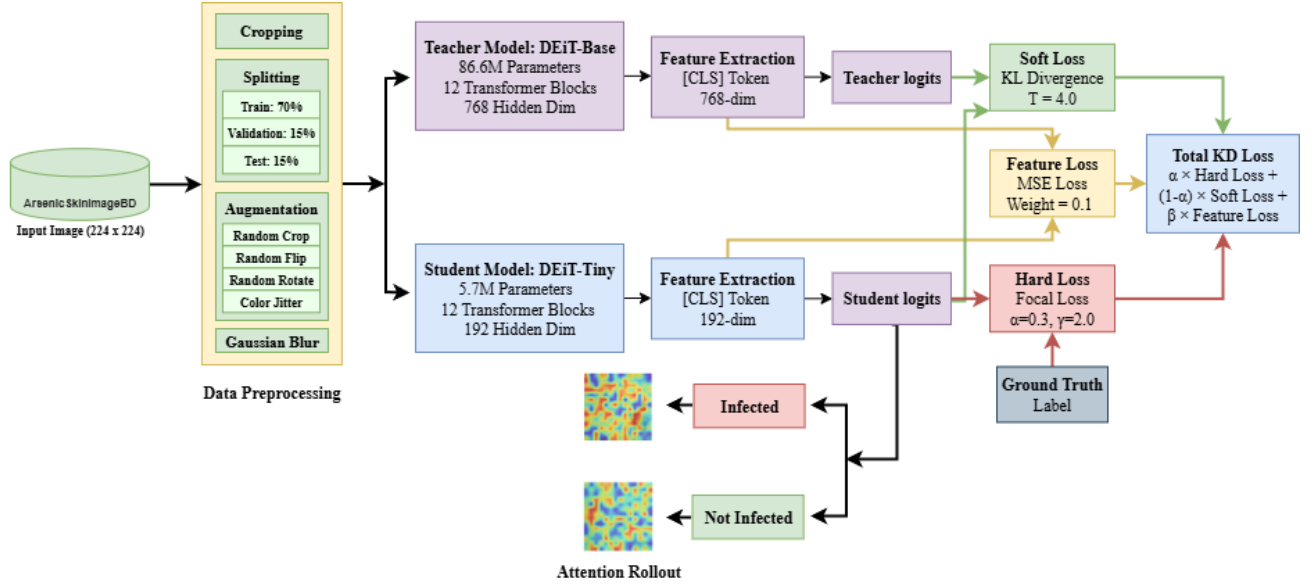


Fig. 1. The Proposed X-ArsenicDistill Framework

training a computationally efficient student model by a large, high-capacity teacher model.

1) *Teacher Model*: DeiT-Base [7] is a robust version of the Vision Transformer that serves the function of the teacher. This model has 86.6 million parameters, and is structured into 12 transformer blocks with a hidden dimension of 768. The teacher model is first fine-tuned on the dataset of arsenic skin lesion with the aim of serving as an expert on this task. It is not intended to be deployed eventually but to produce rich feature representations and soft probability distributions that can be used to provide the so-called “dark knowledge” [6] to drive the learning process of the student.

2) *Student Model*: The model employed is the DeiT-Tiny [7] student model which provides a far smaller and computationally efficient architecture. It has 5.7M parameters, arranged in 12 transformer blocks but with a significantly smaller hidden dimension of 192. It is lightweight and therefore can be used in resource limited environments, including mobile health diagnostic devices. Our main objective in the framework is to make the student model adapt and copy the high performance of the teacher and achieve a trade-off between predictive and computational efficiency.

C. The X-ArsenicDistill Framework

Our methodology is based on the knowledge distillation (KD) process, which is coordinated by a well-designed composite loss function. The total loss guides the student model to receive learning on many inputs, such as the ground-truth labels, softened predictions by the teacher, and its intermediate features representations.

1) *Hard Loss*: The hard loss term, $\mathcal{L}_{\text{hard}}$, grounds the student model’s learning in the ground-truth labels. To address the class imbalance in the ArsenicSkinImageBD dataset, we adopt Focal Loss [15] rather than the standard cross-entropy loss.

Focal Loss reduces the impact of easily classified examples, encouraging the model to concentrate on harder, minority-class samples. It is defined as:

$$\mathcal{L}_{\text{hard}} = -(1 - p_t)^\gamma \log(p_t) \quad (1)$$

Here, p_t represents the predicted probability of the correct class from the student model, and γ is a focusing parameter that adjusts the weighting of well-classified examples.

2) *Soft Loss*: The soft loss, $\mathcal{L}_{\text{soft}}$, is the classic component of knowledge distillation. It encourages the student’s output probability distribution to mimic that of the teacher. This is accomplished by computing the Kullback-Leibler (KL) Divergence between the softened logits of the teacher and student models. A temperature scaling parameter, T , is applied to the logits before the softmax operation to create smoother distributions, which contain richer information about inter-class similarities as perceived by the teacher. The soft loss is formulated as:

$$\mathcal{L}_{\text{soft}} = T^2 \cdot \text{KL}(\sigma(z_s/T) \parallel \sigma(z_t/T)) \quad (2)$$

where z_s and z_t represent the logits produced by the student and teacher models, respectively, $\sigma(\cdot)$ is the softmax function, and T denotes the distillation temperature.

3) *Feature Loss*: To provide a more profound level of guidance, we introduce a feature-based loss, $\mathcal{L}_{\text{feat}}$. This loss encourages the student to learn an intermediate feature space that is similar to the teacher’s. We derive the feature representation out of the final transformer block token [CLS] of each of the two models. There are differences in the feature dimensions, hence a linear projection layer is applied to project the 192-dimensional feature vector of the student into the 768-dimensional space of the teacher. The Mean Squared Error (MSE) is used to compare the projected features of the student

with those of the teacher and thus the distance between their representations is minimised.

4) *Combined Distillation Objective*: The three loss components are linearly combined to form the total distillation objective function, $\mathcal{L}_{\text{TotalKD}}$, which is minimized during training:

$$\mathcal{L}_{\text{TotalKD}} = \alpha \mathcal{L}_{\text{hard}} + (1 - \alpha) \mathcal{L}_{\text{soft}} + \beta \mathcal{L}_{\text{feat}} \quad (3)$$

Here, α is a hyperparameter that balances the influence of the ground-truth labels and the teacher's soft targets, while β controls the strength of the feature-matching regularization. Based on empirical tuning, we set $\alpha = 0.3$ and $\beta = 0.1$.

D. Explainable AI for Model Interpretation

In an effort to promote trust and transparency of our model predictions especially within a sensitive medical application, we include an explainable AI (XAI) technique. Attention Roll-out [8], a technique designed to visualize attention mechanisms in transformer models, is used by us. This method puts the attention maps of all the layers together so as to come up with one detailed visualization that shows those parts in the input image that are most important to the model when it comes to the final classification. The $\mathbf{R}_L$ aggregated attention vector of a model with L layers can be calculated as the result of multiplying recursively attention weight matrices:

$$\mathbf{R}_L = \prod_{l=1}^L (\rho_l A_l + (1 - \rho_l) \mathbf{I}) \quad (4)$$

And where A_l is the attention matrix of layer l , $\mathbf{I}$ is the identity matrix, and ρ_l is a weighting factor (usually equal to 0.5) that represents residual connections. Using this method, we are able to visually examine and contrast the areas of interest of the teacher and the distilled student model, which represents qualitative evidence of effective and meaningful knowledge transfer.

IV. RESULTS AND DISCUSSION

This section is an assessment of our X-ArsenicDistill framework. The narrative includes the description of experimental conditions and then the dynamic of training and quantitative outcomes. Our models are then compared to the SOTA and a qualitative analysis is offered through explainable AI (XAI). Lastly, we comment on the implications and constriction of our results.

A. Experimental Setup

The experiments were run on a Kaggle Notebook with an NVIDIA P100 GPU (16 GB VRAM), 2 Intel Xeon processors and 29 GB of RAM. PyTorch and timm library were used to implement the framework. The optimizer was AdamW [16] based on the Cosine Annealing learning rate scheduler. Table I summarizes the key hyperparameters.

TABLE I
KEY HYPERPARAMETER CONFIGURATION

Parameter	Value
Image Size / Batch Size	224 × 224 / 32
Epochs	20 (early stopping, patience=5)
Optimizer	AdamW (LR: 1e-4)
Loss Components	Focal Loss ($\gamma = 2.0$), KL Div., MSE
KD Parameters	$\alpha = 0.3, \beta = 0.1, T = 4.0$

B. Training Dynamics

The regularizing effect of our distillation framework is indicated in the training histories in Fig. 2. The Student-Scratch model demonstrates the increasing difference between the training and the validation accuracy which is a sign of over-fitting. On the contrary, the validation curve of the Student-KD model follows closely its training curve and shows better generalization. This implies that the guidance of the teacher is effective to ensure that the student does not memorize training set artifacts.

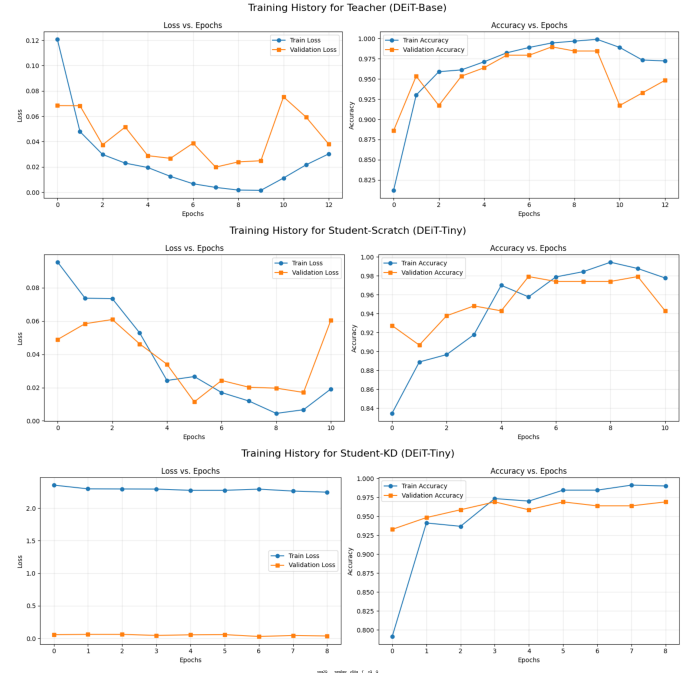


Fig. 2. Training and validation curves for all models.

C. Quantitative Analysis

The confusion matrices of Fig. 3 demonstrates that the Student-KD model decreases false positives by 4 to 1 as compared to the scratch model. This improved accuracy is essential to a quality diagnostic tool. The results of ROC curves (Fig. 4) indicate high levels of discriminative ability of all models, as the AUC scores are 0.99 or above.

Table II summarizes the performance on the held-out test set. The DeiT-Base Teacher sets a strong benchmark with 98.97% accuracy. Our Student-KD model (95.88% acc.) outperforms the baseline Student-Scratch model (95.36% acc.)

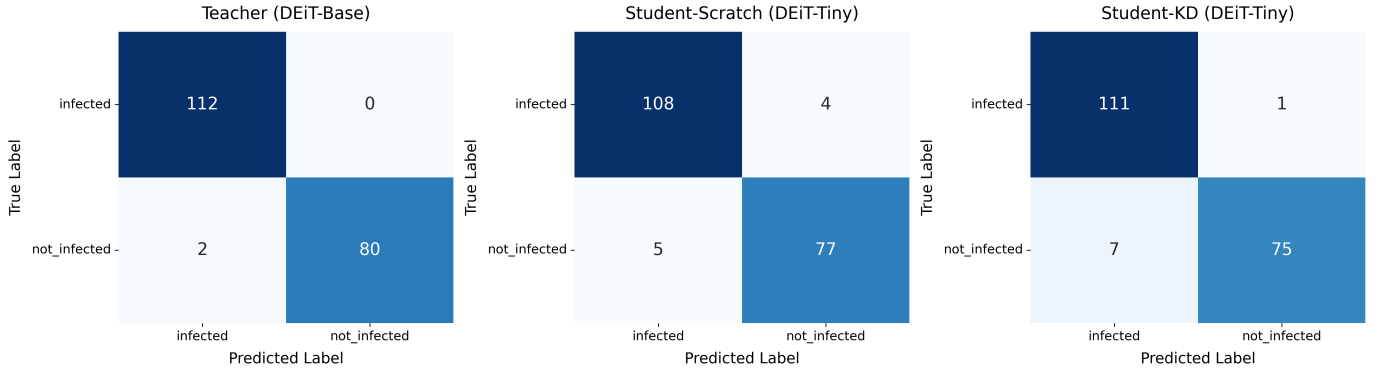


Fig. 3. Confusion matrices for Teacher (left), Student-Scratch (center), and Student-KD (right) models. The distilled student reduces false positives, improving its clinical utility.

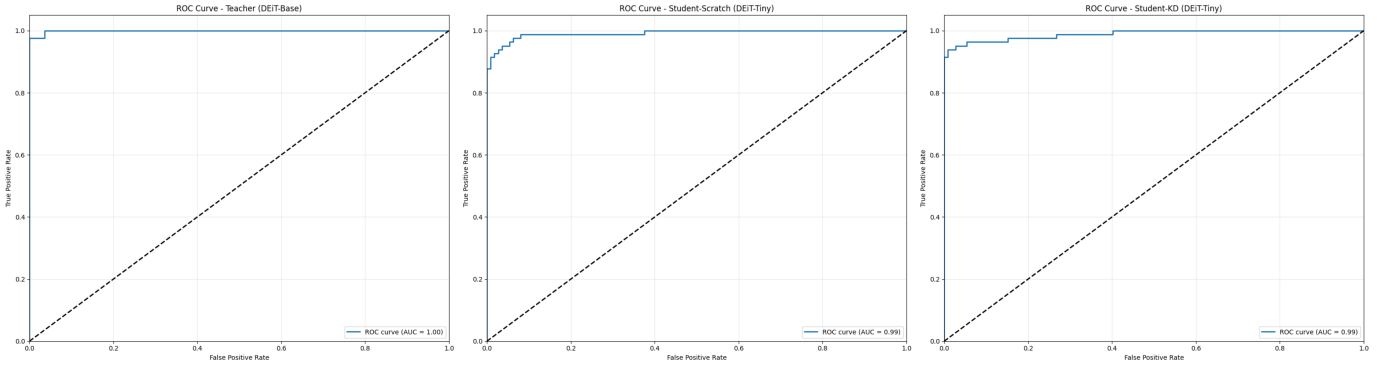


Fig. 4. ROC curves for all models, demonstrating exceptional discriminative capability with AUC scores near unity.

across all metrics, confirming the benefit of distillation. This performance gain is achieved with a **15.5×** parameter reduction compared to the teacher.

TABLE II
PERFORMANCE COMPARISON ON THE TEST SET

Model	Acc	Prec	Rec	F1	Kappa	Params
Teacher	0.9897	0.9912	0.9878	0.9894	0.9788	85.8M
Student-KD	0.9588	0.9638	0.9529	0.9573	0.9147	5.5M

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON
ARSENICSKINIMAGEBD DATASET

Reference	Year	Method	Acc. (%)
Saha et al. [9]	2024	EfficientNet-B1	94.69
Jobayer et al. [10]	2024	Custom CNN	91.00
Sintheia et al. [12]	2025	Custom CNN	98.33
Mehedi et al. [11]	2025	Modified Xception	97.69
Our Teacher	2025	DeiT-Base	98.97
Our Student-KD	2025	KD DeiT-Tiny	95.88

D. Comparison with State-of-the-Art

Table III benchmarks our models against prior works on the ArsenicSkinImageBD dataset. Our DeiT-Base Teacher establishes a new SOTA accuracy of **98.97%**. More importantly, our lightweight Student-KD model is highly competitive, outperforming several larger CNN-based models while being optimized for efficiency.

E. Qualitative Analysis

To solve the problem of "black box" of deep models, we adopt Attention Rollout [8] to visualize model focus. The attention map of the Student-KD model of the case Infected, as in Fig. 5, closely resembles the attention paid to the area of the lesion by the teacher. This is as opposed to the more diffuse attention of the scratch model which implies that our

framework is able to transfer the visual rationale of the teacher, not only its predictive ability. This interpretability is critical in clinical trust.

F. Discussion

Our findings indicate that we have achieved the X-ArsenicDistill framework to fill in the accuracy-efficiency gap. The multi-faceted loss function which is a combination of hard labels, soft probabilities and feature alignment was essential. Not only did it enhance quantitative measures, but also trained the teacher to transfer the localization of the features they learned as it can be seen in the XAI analysis. The main benefit of this holistic knowledge transfer is that it is better than simpler distillation techniques.

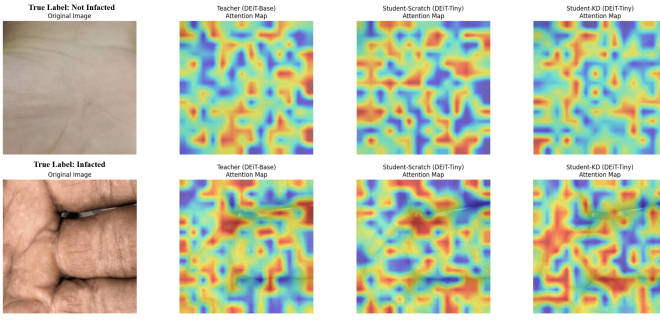


Fig. 5. Attention Rollout visualizations. The Student-KD model's attention more closely aligns with the Teacher's focus on salient regions than the Student-Scratch model.

Clinically, high accuracy and low false positive rate of the Student-KD model make it a more useful tool to screen applications. In addition, the visualization of the attention of the model offers some transparency that is necessary in earning clinicians trust. This great reduction in model size ($15.5\times$) enables the resultant model to be deployed to a mobile or edge device, allowing the access of diagnostic aids in a resource-constrained environment where arsenicosis is common.

One weakness of this study is that it uses a data set on a single geographical location. The work in the future should be aimed at confirming the strength of the model when it is going to be used in a more diverse patient population. Further performance in the efficiency-accuracy trade-off would also be obtained by exploring other lightweight student architectures including MobileViT.

V. CONCLUSION

In this paper, we addressed the critical need for an accurate and computationally efficient model for diagnosing arsenic-induced skin disease by introducing **X-ArsenicDistill**, a novel framework that employs knowledge distillation to transfer knowledge from a large DeiT-Base teacher to a compact DeiT-Tiny student. By utilizing a multi-faceted loss function that combines hard labels, soft targets, and intermediate feature alignment, we successfully trained a lightweight model that achieves a test accuracy of 95.88%, significantly outperforming its baseline counterpart and demonstrating performance competitive with larger SOTA models despite a $15.5\times$ reduction in parameters. Furthermore, our qualitative analysis using Attention Rollout revealed that the student model not only learned to predict accurately but also emulated the teacher's visual rationale by focusing on clinically relevant lesion areas. The X-ArsenicDistill framework thus presents a viable pathway for developing robust, lightweight, and interpretable diagnostic tools. However, we acknowledge the potential for domain shift arising from variations in camera sensors, ethnicities, and skin tones, and how these factors might affect generalization. Consequently, future work will prioritize multi-center data collection and the application of domain adaptation techniques to enhance model robustness across diverse populations.

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